



PAPER – 6:

FINANCIAL MANAGEMENT AND STRATEGIC MANAGEMENT

6A: FINANCIAL MANAGEMENT



QUESTIONS

Division A: Case Scenario

Investment Decision

1. Mr. Ronak, a doctor by profession, has his own private hospital at Goa having specialization in cardiac treatments. However, now-a-days, Goa not only being a place for the tourists, but is also a place for business delegates, cultural people, politicians, students and other classes of people. Gradually, Goa is opening new windows for businesses and getting recognition as an important tourist and leisure hub in South West India.

There are a number of hotels and resorts at Goa. However, the need still exists for more hotel services, in particular with the excellent service, and because of the large number of visitors from all over the country and all walks of life always favour Goa state for their recreation.

Mr. Ronak although being a doctor by profession is contemplating to establish a five-star hotel at Goa. The hotel will consist of 5 floors. The hotel will include 40 normal rooms and 8 deluxe suites, as well as a restaurant and couple of conference rooms with a small wedding hall on the ground floor. Following are the estimated occupancy rate including

fare composition in the Table 1. Being a five-star hotel, breakfast would be complementary but lunch and dinner are on a-la-carte basis.

Table 1: Hotel accommodation, estimated occupancy rate and fare.

Types of Facility	Numbers	Occupancy Rate	Average Rent Per Room Per Day	Growth Rate in Rent
Normal Room	40	33% or 120 Days	₹ 8000	12%
Deluxe Suites	8	33% or 120 Days	₹ 25,000	9%
Conference with Wedding Hall	2	40 days	₹ 3,00,000	9%
Restaurant	1	All days	₹ 27,000 sales per day	8%

For the sake of simplicity in calculation, growth rate to be applied only once after completion of 10 years.

The estimated cost of land will be ₹ 250 million and the construction cost will be ₹ 100 million. The estimated salvage value at the end of 15th year will be 25% of the cost of construction. The cost of furniture will be of ₹ 1,50,000 for each normal room and ₹ 3,80,000 for each deluxe suite. The cost of the furniture for the conference room with wedding hall will be ₹ 7,00,000 each and for restaurant it will be 10,00,000. In addition, the hotel will require 4 elevators at different locations and will be costing around ₹ 35,00,000 each. The cost of buying and installing electronic appliances like TV sets, Air conditioners, Fridge etc. will be around ₹ 35 million. Elevators would be depreciated at a rate of 5% p.a. Electronic appliances will have a salvage value of 15% of its acquisition cost at the end of 15 years.

The hotel will be built by renowned builder named '**Harihar Infrastructure**'. The builder estimated that building will survive for 15 years. The required furniture will be supplied by the local reputed furniture company named **Veru Furnishings Ltd**. They ensured that furniture will go for 10 years very smoothly. At the end of tenth year, new furniture for normal rooms and deluxe suites will be bought and old

furniture for the same will be sold by the hotel owner. The owner of the hotel estimates that he would be able to purchase the required furniture at 15% higher price than the previous purchase price. The salvage values of the furniture at the end of tenth year will be 5% of their purchase prices with no book value remaining. Furniture at restaurant, conference and wedding hall will not require any major changes as such except for minor renovation which will cost ₹ 20,00,000 in total at the end of 12th year. Any scrap generated on account of such renovation will be sold at ₹ 1,75,000.

In order to boost the tourism industry at Goa, the state govt will be granting subsidy of 15% on the initial capex incurred, it will be paid at the time of cost incurred and additional subsidy of 10% on annual revenue expenses for the first 3 years of operation, but will be credited directly in the bank account only at the end of 5th year and the same shall be non-taxable.

The total annual recurring expenses will be ₹ 1,80,00,000/-. It includes salaries to managers, staff and employees, utilities expenses, house keeping and security services' contract, AMC for electronic appliances, restaurant supplies and materials, other miscellaneous expenses, etc.

After the end of 10 years, annual recurring expenses will increase at a rate of 10% which is to be applied once. Furthermore, the hotel authority is determined to provide the best and professional hotel services to the clients by offering training to the employees. They decided to spend ₹ 5,00,000 per year for the purpose of training of the employees.

The hotel project will be entitled to enjoy tax holiday for the first five years after which the corporate tax rate of 25% will also be applied for the hotel. The Cost of equity for the company is 12% and the estimated hurdle rate by considering the structure of capital of the proposed hotel is fixed at 15%.

(Depreciation to be taken on SLM basis and assume 360 days in a year. Ignore depreciation on furniture used in restaurant, conference and wedding hall)

Based on above, please answer to the following MCQs.

- (i) The amount of net initial investment required is:
 - (a) ₹ 41.044 Crores
 - (b) ₹ 34.887 Crores
 - (c) ₹ 6.156 Crores
 - (d) ₹ 40.74 Crores
- (ii) NPV of the project is:
 - (a) ₹ 7.0532 Cr
 - (b) ₹ 8.4029 Cr
 - (c) ₹ 8.4935 Cr
 - (d) ₹ 2.4700 Cr
- (iii) Pay Back period of the project to recover the initial investment is:
 - (a) 5.12 years
 - (b) 12.02 years
 - (c) 11.80 years
 - (d) 4.46 years
- (iv) Estimated Recurring accounting profit/(loss) for first three years are:
 - (a) ₹ 7.0928 Cr p.a
 - (b) ₹ 6.9078 Cr p.a
 - (c) ₹ 6.9937 Cr p.a
 - (d) ₹ 9.6120 Cr p.a
- (v) IRR of the project is:
 - (a) 16.25%
 - (b) 19.39%
 - (c) 15%
 - (d) 12%

Ratio Analysis

2. KT Ltd.'s opening stock was ₹ 2,50,000 and the closing stock was ₹ 3,75,000. Sales during the year were ₹ 13,00,000 and the gross profit ratio was 25% on sales. Average accounts payable are ₹ 80,000. Creditors Turnover Ratio =?
- (a) 13.33
(b) 14.33
(c) 14.44
(d) 13.75

Leverage

3. A firm has sales of ₹ 75,00,000, variable cost of ₹ 42,00,000 and fixed cost of ₹ 6,00,000.
- It has a debt of ₹ 45,00,000 at 9% and equity of ₹ 55,00,000. Does it have favourable financial leverage?
- (a) ROI is less than interest on loan funds and hence it has no favourable financial leverage.
(b) ROI is equal to interest on loan funds and hence it has favourable financial leverage.
(c) ROI is greater than interest on loan funds and hence it has favourable financial leverage.
(d) ROI is greater than interest on loan funds and hence it has unfavourable financial leverage.

Division B: Descriptive Questions

Ratio Analysis

4. Following are the data in respect of LP enterprises for the year ended 31st March, 2024:
- | | | |
|---------------------------------|---|------|
| Debt to Total assets ratio | : | 0.40 |
| Long-term debts to equity ratio | : | 30% |
| Gross profit margin on sales | : | 20% |

Accounts receivables period	:	36 days
Quick ratio	:	0.9
Inventory holding period	:	60 days
Cost of goods sold	:	₹ 64,00,000

Liabilities	₹	Assets	₹
Equity Share Capital	20,00,000	Fixed assets	
Reserves & surplus		Inventories	
Long-term debts		Accounts receivable	
Accounts payable		Cash	
Total	50,00,000	Total	

Required:

COMPLETE the Balance Sheet of LP enterprises as on 31st March, 2024.
All calculations should be in nearest Rupee. Assume 360 days in a year.

Cost of Capital

5. BS Ltd. has the following capital structure at book-value as on 31st March, 2024:

Particulars	(₹)
Equity share capital (10,00,000 shares)	3,00,00,000
11.5% Preference shares	60,00,000
10% Debentures	1,00,00,000
	4,60,00,000

The equity shares of the company are sold for ₹ 300. It is expected that the company will pay next year a dividend of ₹ 15 per equity share, which is expected to grow by 5% p.a. forever. Assume a 35% corporate tax rate.

Required:

- (i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.
- (ii) COMPUTE the new WACC, if the company raises an additional ₹ 50 lakhs debt by issuing 10 years 12% debentures but the yield on debentures of similar maturity and risk class is 13%; flotation cost is 2%. Face value of the debenture is ₹100. This would result in increasing the expected equity dividend to ₹ 20 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 250 per share.

Capital Structure

6. Company XYZ is unlevered and has a cost of equity of 20 percent and a total market value of ₹ 10,00,00,000. Company ABC is identical to XYZ in all respects except that it uses debt finance in its capital structure with a market value of ₹ 4,00,00,000 and a cost of 10 percent. FIND the market value of equity, weighted average cost of capital and cost of equity of ABC if the tax advantage of debt is 25 percent.
7. The following data relate to two companies belonging to the same risk class:

Particulars	A Ltd.	B Ltd.
Expected Net Operating Income	₹ 18,00,000	₹ 18,00,000
12% Debt	₹ 54,00,000	-
Equity Capitalization Rate	-	18

Required:

- (a) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per M.M. Approach.
- (b) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per M.M. Approach.

Leverage

8. Following data of PC Ltd. under Situations 1, 2 and 3 and Financial Plan A and B is given:

Installed Capacity (units)	3,600
Actual Production and Sales (units)	2,400
Selling price per unit (₹)	30
Variable cost per unit (₹)	20
Fixed Costs (₹):	
Situation 1	3,000
Situation 2	6,000
Situation 3	9,000

Capital Structure:

Particulars	Financial Plan	
	A	B
Equity	₹ 15,000	₹ 22,500
Debt	₹ 15,000	₹ 7,500
Cost of Debt	12%	12%

Required:

- CALCULATE the operating leverage and financial leverage.
- FIND out the combinations of operating and financial leverage which give the highest value and the least value.

Dividend Decision

9. The following information is taken from Gamma Ltd.

Net Profit for the year	₹ 30,00,000
12% Preference share capital	₹ 1,00,00,000
Equity share capital (Share of ₹ 10 each)	₹ 60,00,000
Internal rate of return on investment	22%
Cost of Equity Capital	18%

Retention Ratio 75%

CALCULATE the market price of the share using:

- (1) Gordon's Model
- (2) Walter's Model

Working Capital

10. TMT Limited is commencing a new project for manufacture of electric toys. The following cost information has been ascertained for annual production of 60,000 units at full capacity:

		Amount per unit (₹)
Raw materials		20
Direct labour		15
Manufacturing overheads:		
	₹	
Variable	15	
Fixed	<u>10</u>	25
Selling and Distribution overheads:		
	₹	
Variable	3	
Fixed	<u>1</u>	<u>4</u>
Total cost		64
Profit		<u>16</u>
Selling price		<u>80</u>

In the first year of operations expected production and sales are 40,000 units and 35,000 units respectively. To assess the need of working capital, the following additional information is available:

- (i) Stock of Raw materials 3 months consumption.
- (ii) Credit allowable for debtors 1½ months.
- (iii) Credit allowable by creditors 4 months.

- (iv) Lag in payment of wages 1 month.
- (v) Lag in payment of overheads $\frac{1}{2}$ month.
- (vi) Cash in hand and Bank is expected to be ₹ 60,000.
- (vii) Provision for contingencies is required @ 10% of working capital requirement including that provision.

You are required to PREPARE a projected statement of working capital requirement for the first year of operations. Debtors are taken at cost.

11. Banu Limited is considering relaxing its present credit policy and is in the process of evaluating two proposed policies. Currently, the firm has annual credit sales of ₹ 225 lakhs and accounts receivable turnover ratio of 5 times a year. The current level of loss due to bad debts is ₹ 7,50,000. The firm is required to give a return of 20% on the investment in new accounts receivables. Policy option II requires a manager to manage the receivables with salary of ₹ 50,000 per month. The company's variable costs are 60% of the selling price. Given the following information, which is a better option?

(Amount in lakhs)

	Present Policy	Policy Option I	Policy Option II
Annual credit sales (₹)	225	275	350
Accounts receivable turnover ratio	5	4	3
Bad debt losses (₹)	7.5	22.5	47.5

Miscellaneous

12. (a) EXPLAIN the advantages and disadvantages of both profit maximization and wealth maximization goals
- (b) EXPLAIN in brief the features of various types of preference shares.